



Cambridge IGCSE™

BIOLOGY

0610/42

Paper 4 Theory (Extended)

February/March 2026

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2026 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **16** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance
For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	correct point or mark awarded
	incorrect point or mark not awarded
	information missing or insufficient for credit
	allow or accept
	incorrect or insufficient point ignored while marking the rest of the response
	contradiction in response, mark not awarded
	benefit of the doubt given
	error carried forward applied
	point has been noted, but no credit has been given or blank page seen
	correct awarding one mark from marking point or marking group 1. similar numbered ticks are used for marking point or marking groups 2, 3, 4 etc.

Annotation	Meaning
	pages are linked together
	used to highlight part of the response
	used to highlight parts of an extended response
	used to highlight parts of an extended response
	Point already given
	Maximum mark reached
	Reject
	Key point attempted / working towards marking point / incomplete answer / response seen but not credited / blank page seen
	Maximum number of marks for a marking point has been awarded.

Mark Scheme Abbreviations:	
;	separates marking points
/	alternative responses for the same marking point
R	reject the response
A	accept the response
I	ignore the response
ecf	error carried forward
AVP	any valid point
ora	or reverse argument
AW	alternative wording
underline	actual word given must be used by candidate (grammatical variants excepted)
()	the word / phrase in brackets is not required but sets the context
max	indicates the maximum number of marks that can be given
MP	marking point

Question	Answer	Marks	Guidance												
1(a)(i)	<i>any two from:</i> plant ; fungus ; prokaryote ; protocist ;	2	R additional answers on each line if incorrect with the exception of bacteria												
1(a)(ii)	<table><tr><td>their cells have a cell wall</td><td></td></tr><tr><td>they have lungs</td><td></td></tr><tr><td>they are multicellular</td><td>✓ ;</td></tr><tr><td>they are vertebrates</td><td></td></tr><tr><td>they obtain their energy by consuming other organisms</td><td>✓ ;</td></tr><tr><td>they respire</td><td>✓ ;</td></tr></table>	their cells have a cell wall		they have lungs		they are multicellular	✓ ;	they are vertebrates		they obtain their energy by consuming other organisms	✓ ;	they respire	✓ ;	3	R each additional tick 5 ticks and 3 correct = 1 mark 4 ticks and 3 correct = 2 marks 4 ticks and 2 correct = 1 mark
their cells have a cell wall															
they have lungs															
they are multicellular	✓ ;														
they are vertebrates															
they obtain their energy by consuming other organisms	✓ ;														
they respire	✓ ;														
1(b)(i)	X drawn on a base ;	1													
1(b)(ii)	A and C and T ; A pairs with T and C pairs with G ;	2	2nd marking point gets MP1 and MP2												
1(c)(i)	clouded leopard ;	1													
1(c)(ii)	lion and leopard ; most <u>recent common</u> ancestor / most <u>recent</u> branch ;	2	MP2 is dependent on MP1												
1(d)(i)	<i>cell membrane:</i> A ; <i>nucleus:</i> D ;	2													
1(d)(ii)	B ; <u>aerobic</u> respiration ; releases energy ;	3													
1(d)(iii)	<u>protein</u> synthesis / making <u>proteins</u> ;	1													

Question	Answer	Marks	Guidance
1(d)(iv)	image size / AW ;	1	

Question	Answer	Marks	Guidance								
2(a)(i)	12.75 / 12.8 ;;	2	MP1 $85 / 0.85$; MP2 $15(.0) \times (85 \div 100) = 12.75$ or $15(.0) \times 0.85 = 12.75$; A ecf for MP2								
2(a)(ii)	<table><tr><td>At all temperatures, species 3 has a greater rate than species 1.</td><td></td></tr><tr><td>Species 1 has a rate of at least 50% of the maximum over a temperature range of 23 °C.</td><td>✓</td></tr><tr><td>Species 2 has a greater rate of photosynthesis than species 1.</td><td></td></tr><tr><td>Species 3 has a higher optimum temperature than species 1.</td><td>✓</td></tr></table>	At all temperatures, species 3 has a greater rate than species 1.		Species 1 has a rate of at least 50% of the maximum over a temperature range of 23 °C.	✓	Species 2 has a greater rate of photosynthesis than species 1.		Species 3 has a higher optimum temperature than species 1.	✓	2	R each additional tick 3 ticks and 2 correct = 1 mark 3 ticks and 1 correct = 0 marks 4 ticks = 0 marks
At all temperatures, species 3 has a greater rate than species 1.											
Species 1 has a rate of at least 50% of the maximum over a temperature range of 23 °C.	✓										
Species 2 has a greater rate of photosynthesis than species 1.											
Species 3 has a higher optimum temperature than species 1.	✓										

Question	Answer	Marks	Guidance
2(a)(iii)	<i>any six from:</i> 1 (rate) increases and then decreases ; 2 <i>ref to</i> optimum / fastest rate, at 5 °C ; 3 no photosynthesis / rate is 0, above 30 °C ; <i>up to 5°C</i> 4 temperature is the limiting factor ; 5 <u>kinetic</u> energy (of particles) increases / ora; 6 increased <u>frequency</u> of (effective) collisions / ora ; 7 more enzyme-substrate complexes form / ora ; <i>above 5 °C</i> 8 <i>ref to</i> denaturation ; 9 <u>active site</u> changes shape / <u>active site</u> no longer <u>complementary</u> (shape) to substrate ; 10 fewer enzyme-substrate complexes form / less frequent collisions ; 11 at high temperatures stomata close preventing entry of carbon dioxide ;	6	
2(b)(i)	$6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$;;	2	M1 LHS M2 RHS 1 mark for unbalanced photosynthesis formula $x\text{CO}_2 + x\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + x\text{O}_2 = 1 \text{ mark}$ x = any / no, number

Question	Answer	Marks	Guidance								
2(b)(ii)	make <u>chlorophyll</u> ; transfers <u>light (energy)</u> into <u>chemical energy</u> ;	2									
2(c)	<table><tr><td>carbohydrate</td><td>function</td></tr><tr><td>starch</td><td>energy <u>store</u> ;</td></tr><tr><td>cellulose</td><td>(build / component of / AW,) cell walls ;</td></tr><tr><td>sucrose ;</td><td>translocated in the phloem</td></tr></table>	carbohydrate	function	starch	energy <u>store</u> ;	cellulose	(build / component of / AW,) cell walls ;	sucrose ;	translocated in the phloem	3	
carbohydrate	function										
starch	energy <u>store</u> ;										
cellulose	(build / component of / AW,) cell walls ;										
sucrose ;	translocated in the phloem										

Question	Answer	Marks	Guidance
3(a)(i)	J artery and K vein ;	1	both required for the mark
3(a)(ii)	P drawn on left-hand side in vessel J ;	1	

Question	Answer	Marks	Guidance
3(a)(iii)	<i>any five from:</i> 1 form network / (highly) branched / AW ; 2 increased surface area ; 3 <u>walls</u> one cell thick / thin <u>walls</u> ; 4 reduced / small, diffusion <u>distance</u> ; 5 permeable (walls) ; 6 for, <u>exchange</u> / <u>diffusion</u>, of (named) substances ; 7 narrow lumen ; 8 red blood cells pass through slowly ; 9 to get close to <u>cells</u> / penetrate <u>all</u> <u>tissues</u> / supply <u>all</u> the <u>cells</u> / AW ; 10 AVP ;	5	mark independently e.g. plasma able to cross wall and form tissue fluid / white blood cells able to cross wall
3(b)(i)	<i>any two from:</i> red blood cells ; platelets ; plasma ;	2	

Question	Answer	Marks	Guidance
3(b)(ii)	<i>any four from:</i> 1 <u>produce</u> antibodies ; 2 antibodies have complementary (shape) to (specific) antigens ; 3 antibodies bind to, antigens / pathogens ; 4 antibodies destroy pathogens (directly) ; 5 antibodies mark / AW, pathogens for, phagocytosis / destruction by phagocytes ; 6 <u>formation / production</u>, of memory cells ; 7 give, <u>active / long-term</u>, immunity ; 8 AVP ;	4	e.g. antibodies, neutralise toxins / agglutinate or clump pathogens together / lymphocytes divide or clone themselves (by mitosis) when encounter a, pathogen or antigen / immobilise pathogen antibodies immobilise pathogens for destruction by phagocytes = MP5 and MP8

Question	Answer	Marks	Guidance
4(a)(i)	L cornea ; M iris ;	2	
4(a)(ii)	F drawn on fovea ;	1	

Question	Answer	Marks	Guidance
4(a)(iii)	<i>any three from:</i> contains (high density of) cones ; <u>three</u> types of cone cell ; (different cones) <u>absorb light</u> of different, colours / wavelengths ; <u>colour</u> vision / detection of <u>colour</u> / produce <u>coloured</u> image ;	3	
4(b)(i)	<i>any four from:</i> 1 ciliary muscles <u>relax</u> ; 2 suspensory ligaments, tensioned / pulled (tight) / are taut ; 3 lens, pulled flat / flattens / becomes thin(ner) / less convex ; 4 less <u>refraction</u> (of light) / doesn't need to <u>refract</u> (light) as much ; 5 focuses <u>light</u> onto, retina / fovea ;	4	
4(b)(ii)	accommodation ;	1	

Question	Answer	Marks	Guidance
5(a)(i)	<i>male:</i> red and <i>female:</i> white ;	1	
5(a)(ii)	homozygous ;	1	

Question	Answer	Marks	Guidance															
5(a)(iii)	<table border="1"> <tr> <td></td><td></td><td colspan="2">male</td></tr> <tr> <td></td><td></td><td>X^R</td><td>Y</td></tr> <tr> <td rowspan="2">female</td><td>X^r</td><td>$X^R X^r$</td><td>$X^r Y$</td></tr> <tr> <td>X^r</td><td>$X^R X^r$</td><td>$X^r Y$</td></tr> </table> 1 red-eyed female : 1 white-eyed male ;;;			male				X^R	Y	female	X^r	$X^R X^r$	$X^r Y$	X^r	$X^R X^r$	$X^r Y$	3	MP1 for male and female gametes MP2 for offspring genotypes MP3 for phenotypic ratio
		male																
		X^R	Y															
female	X^r	$X^R X^r$	$X^r Y$															
	X^r	$X^R X^r$	$X^r Y$															
5(a)(iv)	sex linkage ;	1																
5(b)	four ; diploid ; mitosis ; replicate ;	4																

Question	Answer	Marks	Guidance
6(a)(i)	<i>producers</i> : hornwort and phytoplankton ; <i>only feeds as secondary consumer</i> : shrimp ; <i>consumes at three different levels</i> : (rainbow) trout ;	3	
6(a)(ii)	phytoplankton → daphnia / copepod → shrimp → newt → rainbow trout → bald eagle ;;	2	MP1 for 6 correct organisms only MP2 arrows indicating correct feeding relationships
6(a)(iii)	moist skin / AVP ;	1	
6(a)(iv)	(population of eagles may) decrease ; fewer rainbow trout to eat / rainbow trout are their only source of food ;	2	
6(a)(v)	<i>any three from</i> : crop plants, are producers / in first trophic level / at a lower trophic level (than rainbow trout) ; shorter food chain (if humans eat crop plants) / humans become a primary consumer instead of a secondary consumer ; energy is lost (at each trophic level) / energy transfer is inefficient ; named example of energy loss ;	3	A ora e.g. respiration / heat / digestion / movement / undigested / not all eaten / faeces / urine